

Interlanguage Research Method, Evidence Graphs, and Manager Handoffs

Current research and reproducibility report, v0.5

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1 Interlanguage Research Method, Evidence Graphs, and Manager Handoffs

1.1 Current release

This release consolidates the current interlanguage research layer used by the mathematical-manuscript translation programme. It reconciles the Fable and ChatGPT Pro research record, reproduces the arithmetic of the current Interslavic State C and unified marker packages, and delivers one common method to eight persistent language-group management lanes.

The full editable research department, executable audit, exact automaton data archives, extracted CSV/JSON/Markdown bodies, and lane-specific handoffs are in 12_Interlanguage_Research_Department_and_Automata_v05_20260717.zip.

1.2 What the current model actually contains

Three mathematical objects have different jobs:

1. A typed, channel-separated evidence graph records provenance and routing from source to concept, form, evidence, and decision.
2. A rooted language-family or reader-cohort tree measures dependence, concentration, branch breadth, and diminishing returns from closely related witnesses.
3. A formal weighted automaton or transducer is appropriate only when an extraction process specifies an alphabet, states, transitions, a semiring, initial/final behavior, and path aggregation.

The preserved data packages historically use the word automaton, but their current operational role is evidence-graph serialization and source routing.

1.3 Reproduced findings

The audit reproduces the Interslavic State C snapshot with 1,229 ledger rows, 1,215 unique term IDs, 100 concept buckets, and 9,124 graph edges. Its branch mass is East/West/South = 2341/223/239, giving effective branch count $D1 = 1.7537043785$. The W0 filtered projection changes the mass to 2341/333/348 and $D1 = 1.993192$, but W0 remains unapplied candidate work.

The unified v6.2 package contains 15 lane rows, 1,522 marker rows, 6,088 edges, and 362 concept rows. Its reported readiness value is exactly the formula $100 * m / (m + a + 1)$, where m combines support and candidate mass and a is adverse mass. Because candidate-only lanes can therefore score above 95, this number is rejected as readiness, correctness, quality, or translation priority.

Both packages' output manifests verify. Neither package pins the complete generation-input chain, and neither arithmetic result certifies linguistic decisions.

1.4 Operational method

Every manager now uses the same evidence rules across every mathematical work:

- start from controlled concepts and intended senses, not English strings;
- keep support, candidate, competitor, adverse, absence, and veto channels separate;
- distinguish concept-shelf evidence from row-verified witnesses;
- treat generated translations as consistency material, never native evidence;
- state the evidence unit, language/cohort partition, dependence model, provenance distribution, and missing data before reporting branch metrics;
- require source, sense, script/register, adverse-evidence, and review gates instead of scalar auto-promotion;
- preserve the pre-change state, applied diff, and unresolved reviewer question;
- reserve external/community status for actual external/community returns.

The eight management lanes are Slavic/Interslavic; Chinese/Japanese/Korean; Spanish/French/Romance; English/Germanic; Turkic; Arabic/Persianate/RTL; Malay/Southeast Asian/Pacific; and Horn of Africa/West Africa. These lanes own translations for every work, while the research layer supplies shared method and evidence controls.

1.5 Status and limitations

This is a complete release of the current research-method package, not a claim that any language or manuscript corpus is complete. No lane currently has an accepted external/community review return. Human comprehension experiments, symmetrically justified family/cohort trees, row-level context verification, latent-pivot audits, and target-specific source expansion remain open.

The release is provided so that the reasoning, data, rejected metrics, scripts, and management handoffs can be inspected and reused rather than remaining inside transient AI sessions.

2 Research authority and provenance

Date: 2026-07-17

2.1 Authority order

Authority is resolved by claim lineage and reconciliation, not by filename rhetoric.

1. **Live Fable programme** — newest synthesis, corrections, measurements, source policy, and July 10 consolidated interface: C:\Users\Floris\Documents\CLAUDE PLEASE DONT DELETE WINDOWS 32_claude_aid\interlingua_program_20260704
2. **ChatGPT Pro research framework and generated passes** — theoretical reconstruction, formal proposals, red-team audit, and later evidence-graph packages: C:\Users\Floris\Documents\CLAUDE PLEASE DONT DELETE WINDOWS 32_claude_aid\interlingua_program_made flr with chat web stuff\ChatGPT-Research Framework Construction.md
3. **Versioned GitHub release** — packaged 2026-07-06 handoff, including v3 State C, multilane v5, and unified v6.2: C:\Users\Floris\Documents\CLAUDE PLEASE DONT DELETE WINDOWS 32_claude_aid\interlingua_program_20260704\user made flr with chat web stuff\INTERLANGUAGE_GITHUB_RELEASE_20260706.zip
4. **Workspace mirror** — convenient live copies of Fable programme files: C:\Users\Floris\Documents\
5. **Earlier practice stacks and AI drafts** — evidence of process and candidate discovery, never automatically upgraded to current authority.

The reduced GitHub methodology handoff at C:\IL_GitHub\01_other_pc_full\workflow\interlanguage-methodology\20260704_claude_interlingual_harmonization_program is not the full live authority.

2.2 Conflict-resolution rule

When artifacts disagree:

1. Preserve the earlier state as archaeology.
2. Identify the input versions, row identity rules, counting unit, and provenance tier.
3. Prefer a later artifact only when it explicitly reconciles the discrepancy or reproduces the authoritative inputs.
4. Keep projections, candidates, and current observations separate.
5. Never let “certified,” “reviewed,” or “authoritative” in an internal package imply community validation.

This rule resolves the main weighting dispute:

Artifact/state	Role now	St
Baseline (2395,64,59), D1=1.257252	Frozen pre-backfill archaeology	Cu
Automaton v2 (2396,167,169), D1≈1.581	State/edge architecture	St
Reconciliation v2.1	Explains old-ID and join drift	Ac
State C v3 (2341,223,239), D1=1.753704	Current serialized post-writeback snapshot	Cu
State C + W0 (2341,333,348), D1=1.993192	Candidate-after-context projection	Pr
State D D1≈2.62	Mixed token and presence units	Wi
Fifteen-concept presence comparison 2.974 -> 2.866	Honest narrow sensitivity result	Va

The v3 Markdown calls State C “current certified measurement.” In this department that phrase is normalized to **current reconciled serialized dataset snapshot**. It does not mean linguistically certified, externally reviewed, or community approved.

2.3 Authorship lineage

Authorship follows the live programme’s AUTHORSHIP.md:

- **Floris**: direction; marginal intelligibility; dense branch-point heuristic; non-dominance requirement; positive/negative weighting intuition; weighted witness-tree seed; East-Slavic skew catch; global-not-Eurocentric requirement; freeze-then-fix rule; global-register weights extension.
- **Claude Fable 5**: corpus synthesis; measurements; ledgers; scripts; source-use enforcement; branch probes; reconciliation review; framework-paper prose.
- **ChatGPT Pro / GPT-5.5**: theoretical reconstruction; ILO, false-friend-risk, and barycenter hypotheses; linkage/witness/review separation; support/adverse schema;

dependence-corrected tree weighting; formal mathematics pass; generated evidence-graph packages; defect predictions.

- **OpenAI Codex lanes:** operational translation corpora, practice stack, terminology records, and original global mathematical-register pilot idea in the publication reflections.
- **Gemini Deep Research:** seed literature reports and bibliography leads. Its scenario document is not an authority layer.
- **Kimi and earlier systems:** historical drafts and candidate material only unless independently promoted through the evidence rules.

Global mathematical-register lineage is: originating Codex session -> Floris's global-weights extension -> Fable's formalization and measurement.

2.4 Evidence classes and permitted use

Evidence class	May normalize source concept?	May propose target
Canonical source-language authority	Yes	
Native independent target-language source	Yes	
Language-family reference shelf	Yes	
Draft translation	Limited	
Generated corpus / internal consistency material	Limited	
AI synthesis or triangulation	Yes, as QA	
Accepted external/community review	Yes	

Provenance levels are ordered but not interchangeable:

discovered < concept_shelf < row_verified < internally_reviewed < externally_reviewed

2.5 Final weighted-object reconciliation

The debate ended with a layered answer:

- The language-family tree is the correct object for genealogical dependence, branch coverage, and diminishing returns.
- The typed evidence graph is the correct object for source-to-decision provenance and separate support/adverse/gap/candidate paths.
- Weighted automata are a legitimate later formalization only for a specified extraction or routing transducer. Calling a graph an automaton does not by itself define automaton semantics.

The final v3 and v6.2 packages therefore remain useful as evidence-graph serializations. Their names are preserved for provenance; their operational type is stated accurately here.

2.6 Known defects and caveats

- FRAMEWORK_DRAFT_ORDERED_v0.4.md prints the baseline effective-branch value once as 1.255; direct recomputation from (2395,64,59) gives 1.2572524317501557.
- State C v3 contains 1,229 rows but only 1,215 unique term IDs; 27 rows participate in duplicate IDs. Collapsing duplicates barely changes the corpus statistic, but row identity must use the stable row_uid.
- State C v3's manifest hashes outputs, not the full input chain, and references /mnt/data/... generation paths. The arithmetic is reproducible from outputs; the build is not end-to-end reproducible from that package alone.
- Most State C West/South gain is concept-shelf provenance, not row verification: 182 concept-shelf rows carry E/W/S mass 379/162/182; the remaining rows carry 1962/61/57.

- Unified v6.2 contains pseudo-concepts such as ALL, marker IDs, blank values, and policy sentences. It is a routing graph, not a clean lexical authority.
- Unified v6.2's readiness_proxy_0_100 is exactly $100*m/(m+a+1)$, where m is support-plus-candidate mass and a is adverse mass. It ignores source class, support-versus-candidate status, gaps, row review, breadth, script policy, external review, and human comprehension. It is rejected as a readiness measure.
- The dependence_pd six-cohort tree is an asymmetric toy sensitivity model: East is collapsed while West and South are subdivided. It must not become a governance weighting.
- The global-attractor probe is mechanical root-prefix clustering with correspondence folding. Its $16/59 = 27.1\%$ result is hypothesis generation, not intelligibility validation.
- The concept-history model's shipped prediction columns scored precision 0.41 and recall 0.37 in the programme self-test. It remains descriptive only.
- No accepted external review return exists for any lane.

2.7 Canonical current interfaces

- Live policy: SOURCE_USE_POLICY.md
- Live claims: CLAIM_LEDGER.md
- Live heuristics: HEURISTIC_REGISTER.md
- Current framework: FRAMEWORK_DRAFT_ORDERED_v0.4.md
- Current concept interface: CONSOLIDATED_INTERLANGUAGE_INTERFACE_v1_20260710.json/.html
- Current Slavic evidence snapshot: INTERSLAVIC_WEIGHTED_AUTOMATON_STATE_C_PLUS_W0_v3_20260705
- Current global connective findings: GLOBAL_CONNECTIVE_TABLE_v1_20260710.md
- This department's normalized claim status: CLAIM_STATUS_REGISTER.json

If any source package disagrees with this department, record the conflict; do not silently overwrite either state.

3 Interlinguistic method synthesis

This is the shared method for every language manager and every work. It governs translation into the manager's languages; it is not a separate translation lane.

3.1 1. Research objective

Construct and maintain translation registers that maximize **marginal intelligibility over the dominant-language baseline** while preserving source meaning, mathematical function, local linguistic legitimacy, script integrity, and an inspectable evidence trail.

A bridge or controlled register is not good because it is globally familiar or resembles the dominant language. It is useful when it buys an additional reader cohort a usable channel without materially degrading service to existing readers.

The final judge is human comprehension. Corpus counts, source attestations, model geometry, form similarity, branch balance, and AI review are instruments and diagnostics.

3.2 2. Decide the intervention before constructing it

Use the family's standards, scripts, and actual mutual intelligibility, not genetic labels alone.

Situation	Default intervention
Related languages, multiple standards, substantial passive intelligibility	Zonal bridge or coordinated
One strong standard already serves the region	Controlled technical register
Closely connected register split mainly by scripts	Script bridge with determinis
High internal diversity or low mutual intelligibility	Local standards plus concept
Insufficient native technical sources	Source-intake plan only

The source floor is lexicographic: no amount of model output or formal scoring compensates for missing native target-language evidence.

Examples in the present programme:

- Slavic and Romance can support zonal comparison and bridge work.
- Arabic and Malay/Indonesian are controlled-register problems, not invitations to invent replacement languages.
- CJK is a local-standard and crosswalk problem, not a pan-CJK language.
- Horn of Africa, West Africa, Pacific, Indigenous, creole, and sign-language work must be partitioned by actual linguistic and community structure rather than administrative convenience.

3.3 3. Start from concepts, not English strings

Every record begins with a controlled concept and sense, then maps source and target realizations.

Required concept layers:

1. **Proof grammar** — therefore, however, let, suppose, follows, is called, remains.
2. **Curriculum mathematics** — ring, field, ideal, module, homomorphism, quotient.
3. **Work-specific historical register** — the concepts and metaphors of Noether, SGA, or another source corpus.

The 67-row C2 spine is the shared comparison minimum, not a universal vocabulary. Each lane must return witnessed, gap, or not_applicable; it must never invent a form merely to fill a cell.

The current consolidated interface contains 216 concepts across 25 language columns. It is a map and comparison surface. Raw cells, candidate cells, competitors, and adverse items retain their tiers.

3.4 4. Preserve evidence channels

For every concept-candidate-language tuple, record these independently:

- support: a permitted source supports the candidate in the intended sense.
- candidate: the form is useful for discovery or context review but is not yet a witness.
- competitor: the language uses a different form for the same concept.
- adverse: the form collides with another sense, register, script behavior, or community constraint.
- absence: the search has not produced evidence either way.
- veto: a typed prohibition such as false sense, do-not-inherit, or external-authority hold.

A competitor is not absence. Adverse evidence never becomes a negative count that enough positive mass can outvote. A collision can veto a candidate regardless of the number of weak hits.

3.5 5. Separate linkage, witness, and review

These statuses do not imply one another:

1. linked_to_concept
2. candidate_after_context_check
3. concept_shelf_witness
4. row_verified_witness
5. internally_reviewed_for_use
6. externally_reviewed_for_use

Generated text, bilingual drafts, dictionary prefix matches, folded spellings, and model suggestions may discover a candidate. They do not witness living usage.

An internal AI agreement can raise confidence that a record is correctly processed. It cannot set status 6.

3.6 6. Evidence collection loop

For each work and target language:

1. Freeze the current source and target state, including hashes and build identity.
2. Segment the source by stable work, section, and unit IDs.
3. Link each technical term and proof connective to a controlled concept and intended sense.
4. Search native sources, retaining context windows and bibliographic provenance.
5. Classify each return by source class and permitted use.
6. Run sense, homograph, false-friend, script, and register checks.
7. Record support, competitors, adverse evidence, and explicit gaps separately.
8. Compute branch diagnostics only after declaring the unit of evidence and target distribution.
9. Draft a candidate decision and reviewer question; do not auto-promote.
10. Apply accepted internal decisions to translations with source-fidelity, terminology, script, and build gates.
11. Preserve the pre-change state and the diff.
12. Submit held governance questions for appropriate external/community review.

This loop repeats for Noether, SGA, and every other work. A manager's corpus and word web are shared infrastructure for all of its translations.

3.7 7. Sense and adverse-evidence controls

Mechanical lexical search is unsafe without context.

Standing F14 trap concepts:

- modulus: often a module/modulo collision.
- ground form: often generic "fundamental/basic form" or differential geometry.
- complete system: often ODE, residue-system, or generic completeness.
- binary form: often binary code or neural-computation material.
- covariant / contravariant: often tensor or category-theory senses rather than invariant theory.

Standing homograph control: a pan-branch form requires a cross-branch sense check. The Polish jednak "however" versus Croatian/Serbian jednak "equal" case is the canonical warning.

Standing normalization control: diacritic-folded variants may assist search but may not be summed as independent evidence.

3.8 8. Branch, dominance, and fairness controls

Declare what one unit means: document, source, language, cohort, concept-language presence, or row-verified witness. Never mix units in one statistic.

Use raw branch mass to expose the recorded evidence distribution. Use saturation within a language to stop repeated sources from masquerading as independent languages. Use family-tree edge coverage or a declared dependence model to measure marginal breadth.

Keep these questions separate:

- How much permitted evidence exists?
- How many independent branches or cohorts does it cover?
- How far does its distribution deviate from the declared fairness target?
- Which cohorts gain over the dominant baseline?
- What adverse or governance constraints remain?

Population weights are sensitivity analyses unless a community governance target explicitly adopts them.

3.9 9. Scripts and registers are first-class objects

Each lane declares:

- canonical target scripts and encodings;
- deterministic conversions, if any;
- invariants that conversion must preserve;
- register level and audience;
- formula/prose boundary;
- treatment of names, citations, quotations, and source-language fragments;
- punctuation, directionality, and line-breaking behavior;
- build and render checks.

In a dual-script lane, the second script is not a casual transliteration. It is a derived artifact with synchronization gates.

In RTL lanes, bidi behavior, mirrored punctuation, formula direction, embedded Latin identifiers, and citation layout are validation targets.

3.10 10. The global mathematical register

Use it as a comparator column, not a new manager and not an English disguise.

It has three layers:

1. Formula and notation: already substantially global.
2. Proof prose: a small controlled register, but highly branch-divergent in connectives.
3. Concept terms: partly international and measurable per concept.

The current mechanical attractor probe finds 16 of 59 tested concepts in a folded cross-family basin. That is a candidate-generation signal only. The 15-connective pilot finds no cross-family lexical attractor; formula substitutes such as $\exists$, $\therefore$, $\Rightarrow$, and $\stackrel{\text{def}}{=}$ should be shown first where mathematically appropriate.

The operative comparison is:

Does the zonal or local form produce more marginal intelligibility for the target readers than the global/international candidate, after dominance and adverse-evidence controls?

English is always a dominance-bearing carrier in this comparison, never a neutral default.

3.11 11. Model geometry and theoretical research

The ChatGPT Pro reconstruction provides a serious research programme, not current translation authority.

- **Interlingual Local Overlap (ILO)**: test whether aligned concepts preserve local semantic neighborhoods across languages and whether that predicts comprehension.
- **False-friend risk**: test the hypothesis that high form similarity combined with semantic distance predicts errors.
- **Regularized family barycenter**: model a bridge as a family-central solution with learnability and anti-dominance constraints, not as reconstruction of a proto-language.
- **Intrinsic dimension, persistent topology, and bridge complexes**: future comparative instruments only.
- **Latent-pivot audit**: mandatory before using multilingual model geometry, because model spaces may be English-shaped.

None of these model-geometry measurements has been run as validation on the present translation corpus. They may propose experiments and candidates; they cannot currently promote terms.

3.12 12. Decision output

Every consequential term or register decision should produce:

- source and work identity;
- controlled concept and intended sense;
- target language, script, and register;
- candidate forms and variants;
- evidence records by channel and provenance;
- declared branch/cohort model and diagnostic values;
- dominant-baseline comparison;
- adverse and veto controls;
- decision status and rationale;
- unresolved reviewer question;
- invariants and validation gates;
- pre-change state and applied diff, if any.

Use `OPERATIONAL_DECISION_INTERFACE.schema.json` for the machine-readable form.

3.13 13. Promotion rule

No term is promoted because one scalar is high. A production decision requires all applicable hard gates:

- source floor met;
- intended sense verified in context;
- target-language evidence permitted for the claimed use;
- competitors and adverse evidence handled;
- dominance and branch/cohort effects reported;
- script/register invariants defined;
- translation/source fidelity gates pass;
- internal review completed;
- external review or human-comprehension evidence present when the claim requires community legitimacy or intelligibility.

Until then, the correct output is a candidate, a held decision, a gap, or a reviewer question.

4 Mathematical objects and audit

Date: 2026-07-17

4.1 Executive determination

The final research stack contains useful mathematics, but several packages use “automaton” loosely. The operative model is:

typed evidence DAG for provenance and routing
+
rooted family/cohort tree for dependence and breadth
+
multi-objective constrained decision ledger

A formal weighted automaton is a possible implementation of extraction or routing, not the current linguistic evidence object merely by virtue of having weighted edges.

4.2 1. Typed evidence graph

The final State C v3 shape is:

```

START -> SOURCE -> CONCEPT -> FORM
      -> SUPPORT | CANDIDATE | COMPETITOR | ADVERSE | GAP
      -> DECISION

```

This is valuable because it retains typed paths, provenance, and channel separation. It supports queries such as:

- Which source and language support this form?
- Is the evidence concept-shelf or row-verified?
- Which branch provides a competitor?
- Which gap is collection work rather than adverse evidence?
- Which decision is held for authority review?

The final v3 weights are raw branch masses or permitted-use values, not calibrated probabilities. The graph is therefore an evidence-routing serialization, not a probabilistic truth machine.

4.3 2. Formal weighted automata

A decision-grade claim that an object is a weighted automaton must state at least:

- a finite state set Q ;
- an input alphabet Σ (or ranked alphabet for a tree automaton);
- initial and final weights/states;
- a transition relation or transition-weight function;
- a semiring $(K, +, \times, 0, 1)$;
- how path weights are multiplied and alternative paths summed;
- the recognized behavior or transduction;
- the interpretation of acceptance for this task.

The current packages name states and emit edges but do not define a meaningful input language or recognized behavior. They are automaton-shaped graphs. If a later extraction system consumes source tokens or document events and transduces them into typed evidence records, the weighted-transducer formalism becomes exact and useful.

Mohri's finite-state work provides the relevant formal basis for weighted transducers in language and speech processing, and his semiring framework covers weighted directed-graph path algorithms. Those results validate the possible implementation class; they do not validate any linguistic score automatically.

4.4 3. Branch concentration

For declared nonnegative branch masses m_i , let:

$$p_i = m_i / \sum_j m_j.$$

The order-1 effective branch count is:

$$D1 = \exp(-\sum_i p_i \ln p_i).$$

The order-2 count is:

$$D2 = 1 / \sum_i p_i^2.$$

For a declared target distribution π , divergence is:

$$KL(p || \pi) = \sum_i p_i \ln(p_i / \pi_i).$$

When π is uniform over three branches, $KL = \ln(3) - \ln(D1)$. These are concentration statistics. They do not measure intelligibility, evidence quality, or correctness.

4.5 4. Dependence and family-tree coverage

Raw source counts overstate repeated evidence from related languages or duplicated corpora. Let T be a rooted family/cohort tree with nonnegative edge lengths λ_e , and let A be

the set of witnessed leaves. Edge-coverage phylogenetic diversity is:

$$PD_T(A) = \sum_e \lambda_e \mathbb{1}[\text{Desc}(e) \cap A \neq \emptyset].$$

The marginal value of adding leaf l is the total length of its still-uncovered root path. This captures the intended rule: adding another closely related witness usually buys less breadth than adding the first witness from an unrepresented branch.

The positive edge-coverage function is monotone submodular. This supports greedy prioritization of source acquisition under a stated budget. It does **not** make the full decision objective submodular: ambiguity, collisions, script confusion, and dominance constraints can interact and remain separate vetoes or penalties.

Faith-style phylogenetic diversity is total covered tree length. Fair-proportion and equal-splits indices are different leaf-allocation views; they can disagree. Use edge coverage for marginal source breadth and declared leaf weights only for a specified reporting or governance purpose.

The current dependence_{pd} sensitivity calculation is not yet a defensible governance model. It gives East one collapsed cohort while subdividing West and South, so a single East cohort covers less tree length than a single West/South subgroup. A production model must use symmetric resolution or explicitly justify unequal topology and branch lengths.

4.6 5. Candidate selection versus evidence concentration

Do not confuse two objectives:

- **Evidence concentration** asks how skewed the current support is.
- **Candidate/access selection** asks which intervention adds the most usable access under constraints.

For source or review prioritization, a positive maximum-coverage objective can be written:

$$F(S) = \sum_u w_u \max_{\{a \in S\}} q_a(u),$$

where u is an access task and a is a candidate intervention. This benefit layer is a standard monotone submodular form. Adverse evidence, dominance, ambiguity, script failures, and governance rules remain constraints or vetoes.

Hill D1 and KL never choose the best term. A perfectly balanced set of weak or false-sense hits is still bad evidence.

4.7 6. Reproduced State C results

The final v3 term ledger contains:

- 1,229 rows;
- 1,215 unique term IDs;
- 27 rows participating in 13 duplicated IDs;
- 100 concept buckets;
- 9,124 graph edges.

State C current snapshot:

Quantity	Value
East mass	2341
West mass	223
South mass	239
Total	2803
East share	0.8351765965
West share	0.0795576168
South share	0.0852657867
D1	1.7537043785
KL to equal three-branch target	0.5368819503

Quantity	Value
----------	-------

W0 filtered candidate projection:

Quantity	Value
East mass	2341
West mass	333
South mass	348
Total	3022
D1	1.993192
KL to equal three-branch target	0.408875

W0 is a projection only. It must not overwrite State C until its source-row context checks and writeback rules are applied.

Duplicate sensitivity is negligible at corpus level: collapsing repeated term IDs by maximum branch mass changes the snapshot approximately to (2335, 223, 239) and $D1 \approx 1.75514$. That does not excuse the duplicate IDs; stable row identity remains mandatory.

4.8 7. Provenance audit of State C

The branch improvement is not uniformly row-verified:

Provenance group	Rows	East	West	South
concept_shelf writeback	182	379	162	182
blank/unwritten writeback level	1,047	1962	61	57

This means most West/South gain is attributable to concept-shelf coverage. State C is the correct current serialized dataset, but its branch number must be described as a corpus evidence snapshot, not as fully row-verified attestation.

4.9 8. Rejected and withdrawn numbers

- Automaton v2's (2396, 167, 169), $D1 \approx 1.581$ came from stale inputs and old ID joins. Keep its architecture, not its measurement.
- The claimed State D $D1 \approx 2.62$ mixed raw token counts with concept-language presence units. It is withdrawn.
- The honest 15-concept presence-unit comparison changes from (30, 41, 39), $D1 \approx 2.974$ to (39, 44, 76), $D1 \approx 2.866$. More evidence reduced balance because South gained disproportionately. This is a useful warning that “more sources” and “more balance” are not synonyms.
- The early baseline (2395, 64, 59) recomputes to $D1 = 1.2572524318$ and $KL = 0.8696835584$. A 1.255 occurrence in framework v0.4 is a rounding/transcription error.

4.10 9. Unified v6.2 readiness audit

For every nonzero row, the package's `readiness_proxy_0_100` is reproduced by:

$100 * \text{support_candidate_mass} / (\text{support_candidate_mass} + \text{adverse_mass} + 1)$.

This causes lanes with zero support and only candidates to score above 95. It also ignores gaps, evidence provenance, branch breadth, context review, script policy, external review, and human comprehension.

Disposition:

- Preserve it as archaeology under the accurate name `evidence_mass_saturation_proxy` if needed for debugging.
- Reject it as readiness, quality, correctness, or translation priority.
- Replace readiness with a vector of hard gates and unresolved conditions.

The unified graph also includes blank concepts, IDs, and policy sentences as concepts. It is suitable for routing and source discovery after cleaning, not as direct lexical authority.

4.11 10. Other current mathematical claims

Object	Current status	Permitted use
ILO / neighborhood overlap	Proposed, not run on this corpus	Experiment design
FFRisk = FormSim × SemDist	Testable proxy, not validated	Candidate risk ran
Regularized family barycenter	Theoretical model	Comparative candi
Persistent topology / cobordism	Proposed, currently unimplemented	Research only afte
Global-attractor root-prefix clustering	Exploratory mechanical proxy	Candidate discover
History-class prediction	Weak; precision 0.41, recall 0.37	Description only
Hill D1/D2 and KL	Established arithmetic	Evidence-concentr
Edge-coverage PD	Established set function on a declared tree	Dependence-sensit
Human comprehension tests	Not yet run	Required outcome

4.12 11. Minimum mathematical declaration per lane

Before reporting a weight or balance number, a lane must publish:

- evidence unit;
- included and excluded source classes;
- deduplication key;
- language/cohort/branch partition;
- tree topology and edge lengths, if used;
- target distribution and its governance rationale;
- saturation or dependence rule;
- channel treatment;
- provenance distribution;
- sensitivity analysis;
- exact formula and code version;
- known missing data;
- statement that the statistic does not certify intelligibility or correctness.

If those fields are absent, the number is exploratory by definition.

4.13 12. Primary mathematical and empirical anchors

- Mehryar Mohri, *Finite-State Transducers in Language and Speech Processing*: <https://aclanthology.org/J97-2003/>
- Mehryar Mohri, *Semiring Frameworks and Algorithms for Shortest-Distance Problems*: https://jalc.de/issues/2002/issue_7_3/abs-321.pdf
- Anne Chao, Chun-Huo Chiu, and Lou Jost, *Phylogenetic diversity measures based on Hill numbers*: <https://pmc.ncbi.nlm.nih.gov/articles/PMC2982003/>
- Daniel Faith, *Conservation evaluation and phylogenetic diversity*: <https://ricottalab.com/wp-content/uploads/2015/10/faith-1992.pdf>
- Kristina Wicke and Mike Steel, *Combinatorial properties of phylogenetic diversity indices*: <https://arxiv.org/abs/1902.02463>
- George Nemhauser and Laurence Wolsey, *Best Algorithms for Approximating the Maximum of a Submodular Set Function*: <https://pubsonline.informs.org/doi/abs/10.1287/moor.3.3.177>
- Charlotte Gooskens et al., *Mutual intelligibility between closely related languages in Europe*: <https://research.rug.nl/en/publications/mutual-intelligibility-between-closely-related-language-in-europe/>

- Charlotte Gooskens and Vincent van Heuven, *How well can intelligibility of closely related languages in Europe be predicted by linguistic and non-linguistic variables?*: <https://research.rug.nl/nl/publications/how-well-can-intelligibility-of-closely-related-languages-in-euro/>

These sources support the mathematical and measurement families. They do not validate this programme's unpublished weights or linguistic decisions.

5 Research-layer delivery manifest

Generated 2026-07-17.

5.1 Core

- README.md — entry point and final reconciliation.
- RESEARCH_AUTHORITY_AND_PROVENANCE.md — authority order, conflict resolution, authorship, evidence permissions, known defects.
- INTERLINGUISTIC_METHOD_SYNTHESIS.md — shared operational linguistic method for every manager and work.
- MATHEMATICAL_OBJECTS_AND_AUDIT.md — graph/tree/automaton separation, equations, reproduced results, rejected proxies.
- CLAIM_STATUS_REGISTER.json — machine-readable statuses for 27 current claims.
- OPERATIONAL_DECISION_INTERFACE.schema.json — decision-record schema with channel, provenance, gate, branch, risk, and invariant requirements.
- OPERATIONAL_DECISION_INTERFACE.example.json — held ring-row example; intentionally incomplete evidence arrays.
- UNRESOLVED_RESEARCH_QUESTIONS.md — validation, model, lane, and interface gaps.

5.2 Reproducibility

- tools/audit_existing_packages.py
- audit_outputs/AUDIT_RESULTS.json

The audit passed all four hard checks:

- v3 expected values reproduced;
- v3 output manifest verified;
- unified readiness formula reproduced on every lane;
- unified output manifest verified.

5.3 Stable source packages added to workspace

- ..\weighted_automaton\INTERSLAVIC_WEIGHTED_AUTOMATON_STATE_C_PLUS_W0_v3_20260705.zip
- SHA256 3448038E4499DA6CDD8741EE5175076EE19FD6A33023171C641C53F1C6801269
- ..\weighted_automaton\UNIFIED_MARKER_AUTOMATON_v6_2_20260706.zip
- SHA256 4672CAD9DF8C267830321759AB8830B1E6EF000DDA3FE695E83DA063D0D4592F

Both exact ZIPs are extracted beside the archives for inspection.

5.4 Manager handoffs

- LANE_HANDOFFS/SLAVIC_INTERSLAVIC.md
- LANE_HANDOFFS/CJK.md
- LANE_HANDOFFS/ROMANCE.md
- LANE_HANDOFFS/ENGLISH_GERMANIC.md
- LANE_HANDOFFS/TURKIC.md
- LANE_HANDOFFS/ARABIC_PERSIANATE_RTL.md
- LANE_HANDOFFS/MALAY_SEA_PACIFIC.md
- LANE_HANDOFFS/AFRICA_HORN_WEST.md

5.5 Status

- Research synthesis and local artifacts: complete.
- Delivery to all eight existing persistent manager tasks: complete.
- Acknowledgement with current application and conflict report from all eight managers: complete.
- Receipt record: `MANAGER_DELIVERY_RECEIPTS.md`.
- New manager tasks created: zero.

6 Manager delivery receipts

Delivered and acknowledged 2026-07-17. No new tasks were created.

6.1 Slavic / Interslavic

- Task: 019f70bf-07e1-7901-9587-906148002e48
- Handoff read: `LANE_HANDOFFS\SLAVIC_INTERSLAVIC.md` plus the mandatory authority/method set.
- Applied to: Noether corpus reconstruction, including the omitted-but-completed Post-44 payload and closed Fable tranche, before opening SGA work in the same manager.
- Conflicts found: older records overstate canonical/certified status; scalar snapshots were treated too much like decision rules; the hub omits surviving Post-44 payload.
- Resolution acknowledged: State C is the reconciled serialized snapshot; W0 is unapplied projection; no external certification exists; use evidence graph plus declared family/cohort tree.

6.2 Chinese / Japanese / Korean

- Task: 019f70bf-2024-76f3-af4a-7fbfb32a317b
- Handoff read: `LANE_HANDOFFS\CJK.md` plus the mandatory authority/method set.
- Applied to: Noether Paper 36 four-output tranche and Chinese/Japanese reader audit against German R822.
- Conflicts found: historical `SOURCE_BODIES.csv` misclassifies generated/audit material; no pan-CJK or readiness claim is justified; Korean terminology remains provisional.
- Resolution acknowledged: `manifest.csv` controls source classification; maintain separate zh, ja, and ko records connected only by crosswalks.

6.3 Spanish / French / Romance

- Task: 019f70c0-9281-7543-987f-15baabb53a3f
- Handoff read: `LANE_HANDOFFS\ROMANCE.md` plus the mandatory authority/method set.
- Applied to: disk-wide Romance consolidation across Noether ES/FR, SGA 5/6, EGA, Deligne, native examples, and first source-checked tranche.
- Conflicts found: no scope conflict; the old structure permitted flat/scalar ranking and blurred French source/target/register roles.
- Resolution acknowledged: typed evidence graph, declared family/cohort tree, separate French roles, v6.2 98.5 rejected as readiness.

6.4 English / Germanic

- Task: 019f70c0-aa55-7723-b00a-1d95324af359
- Handoff read: `LANE_HANDOFFS\ENGLISH_GERMANIC.md` plus the mandatory authority/method set.
- Applied to: R823 Noether English rebase, current Paper 29 and pending paper deltas, SGA 5/6 synchronization, and shared English/Germanic infrastructure.
- Conflicts found: two SGA-specific tasks already existed before the corrected management model; Noether similarity/delta scores could look decision-like.

- Resolution acknowledged: old SGA tasks are bounded worker-return sources only; no more per-work tasks; similarity scores are triage; publication uses hard-vector gates.

6.5 Turkic

- Task: 019f70c0-d075-7541-943c-e4bd195f9b26
- Handoff read: LANE_HANDOFFS\TURKIC.md plus the mandatory authority/method set.
- Applied to: existing Turkic manager consolidation of source packages, blocker/recovery ledgers, and recovered Noether/Hefferon review material.
- Conflicts found: older Pan-Turkic artifacts contain automaton, marginal-intelligibility, branch-weight, and readiness rhetoric stronger than their evidence.
- Resolution acknowledged: preserve these as historical/noncanonical; no scalar term/readiness decision; current blockers and withheld weights remain compatible with the handoff.

6.6 Arabic / Persianate / RTL

- Task: 019f70c0-dba6-7dd2-bd60-795884fc50f6
- Handoff read: LANE_HANDOFFS\ARABIC_PERSIANATE_RTL.md plus the shared reconciliation.
- Applied to: Noether Paper 06 opening tranche, Arabic and fa_IR.
- Conflicts found: a draft route table listed possible RTL routes too broadly before selected targets were established.
- Resolution acknowledged: unselected routes remain unactivated, not commitments; add typed evidence graph, declared cohort structure, F14 adverse ledger, and explicit bidi invariants.

6.7 Malay / Southeast Asian / Pacific

- Task: 019f70c0-e8d9-7d50-afa5-fa50c6a2600c
- Handoff read: LANE_HANDOFFS\MALAY_SEA_PACIFIC.md plus the mandatory authority/method set.
- Applied to: forensic reconstruction of the old R7 lane across the local hub, GitHub branch, source-body corpus, and uploader packages before the first real translated tranche.
- Conflicts found: historical labels such as `source_gated_draft_support_body` look stronger than current provenance; 59-row and 60-candidate counts are different scopes.
- Resolution acknowledged: normalize to candidate unless exact-sense row-verified native evidence exists; reject 96.1 readiness; never inherit Malay/Indonesian evidence into unrelated SEA/Pacific targets.

6.8 Horn of Africa / West Africa

- Task: 019f70c0-f836-70d1-8d75-b3e9ea86e8d5
- Handoff read: LANE_HANDOFFS\AFRICA_HORN_WEST.md plus the mandatory authority/method sequence.
- Applied to: current GitHub/Zenodo achieved-state audit and then standing translation work across assigned targets.
- Conflicts found: inherited R9 machinery treated the geographic manager too much like one branch and used scalar readiness language.
- Resolution acknowledged: explicit linguistic family/cohort and script/register sublanes; published artifacts establish provenance/routing, not completed translation, promotion, native acceptance, or v6.2 readiness.

6.9 Cross-manager receipt result

All eight tasks accepted the same corrections:

- one manager per language grouping, responsible for every work;
- existing accidental per-work tasks are worker-return sources, not ownership lanes;
- typed evidence graph for provenance and routing;
- declared family/cohort tree only where dependence or breadth is measured;
- no scalar auto-decisions;
- W0 projection is not current accepted writeback;
- unified v6.2 readiness is rejected;
- no community/external certification is claimed.

7 Unresolved research questions

These are research gaps, not reasons to stop ordinary translation work that already has adequate source and QA infrastructure.

7.1 Validation gaps

1. No lane has an accepted external/community review return.
2. No controlled human comprehension experiment has validated the bridge or register choices.
3. Existing intelligibility claims need speaker-language, prior-exposure, education, script, and register controls.
4. Internal AI triangulation needs continued error accounting, but it cannot substitute for points 1–3.

7.2 Evidence-model gaps

1. Define a symmetric or explicitly justified family/cohort topology and branch lengths for each lane that uses dependence correction.
2. Define the evidence unit per calculation and prohibit mixed-unit arithmetic in code and schema validation.
3. Calibrate within-language/source saturation rather than assigning arbitrary “quality” products.
4. Separate provenance rank from numerical permitted-use weight.
5. Complete row-level context verification for concept-shelf writebacks.
6. Establish source-floor thresholds per target without making machine readability a gate against PDF-heavy or under-resourced traditions.
7. Define community-approved target distributions before any governance weighting.

7.3 Formal-model gaps

1. If weighted automata are retained as more than a package name, define the input alphabet, transition behavior, semiring, initial/final weights, accepted behavior, and a task whose path semantics add value.
2. Decide whether extraction is best modeled as a finite-state transducer, a workflow DAG, a probabilistic graphical model, or a database provenance algebra; do not force one formalism across unrelated jobs.
3. Test whether edge-coverage PD improves source-acquisition decisions over simpler branch-first heuristics.
4. Specify uncertainty intervals or sensitivity bands for D1/KL when witnesses and tree structure are incomplete.
5. Clean the unified graph’s pseudo-concepts before it becomes an API source.

7.4 Linguistic-model gaps

1. Run ILO or related neighborhood-overlap measurements on controlled concept/context sets.
2. Audit latent-English/pivot bias before using multilingual-model geometry.

3. Validate $\text{FFRisk} = \text{FormSim} \times \text{SemDist}$ against human error and reaction-time data.
4. Compare regularized-barycenter candidates with actual community and comprehension outcomes.
5. Determine when a global mathematical attractor beats a zonal form and when it merely reflects English/Latin dominance.
6. Complete non-Slavic proof-connective probes; the current 15-row connective result is not globally exhaustive.

7.5 Lane-specific gaps

- **Slavic:** row-verify the long tail; resolve ring, *ređ*, *jednako*, and *važiti* through appropriate authority; complete symmetric dependence modeling; finish dual-script production gates.
- **CJK:** clean source manifests; expand language-specific native source bodies; formalize Simplified/Traditional, Hanja, and typography policies; retain local-standard status.
- **Romance:** fill specialist Noether gaps from specialist sources; test Spanish/French dominance; expand beyond current ES/FR-heavy evidence.
- **English/Germanic:** build independent non-English Germanic witness shelves; prevent English from acting as a hidden pivot for other managers.
- **Turkic:** execute format-relaxed source intake; define subbranches and script policies; test whether any bridge intervention is warranted.
- **Arabic/Persianate/RTL:** expand separate living-language evidence; finish RTL build invariants; retain false-sense traps.
- **Malay/SEA/Pacific:** separate unrelated sublanes; upgrade candidates to target-specific witnesses; avoid administrative pan-language inference.
- **Horn/West Africa:** build the target inventory and source floors from scratch; partition by actual families, standards, and scripts; develop community review paths.

7.6 Interface gaps

1. Add automated JSON Schema validation to manager return pipelines.
2. Add source-hash and context-anchor verification.
3. Add a hard check that generated/draft sources cannot set `target_witness` or `external_certification` uses.
4. Add a hard check that a numeric branch statistic cannot be emitted without unit, partition, target distribution, and interpretation limit.
5. Add pre-change snapshot and diff verification to every applied terminology patch.
6. Define a shared concept-ID registry without making English glosses the identity key.

7.7 Stop conditions for research claims

Stop and report a gap rather than promote when:

- the source sense is unresolved;
- a trap or homograph has no context window;
- support and competitors are collapsed;
- the evidence unit or deduplication key is unknown;
- the lane's source floor is not met;
- the script/register invariants are undefined;
- an external/community claim is requested without external/community evidence;
- a scalar is being used to bypass hard gates.